

A Scaling–Cancellation Principle for Chord-Based Geometric Approximations

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Abstract

This note separates two geometric error-suppression mechanisms that arise in chord-based approximation and angular refinement. The first is scaling–cancellation: when fixed-resolution approximants satisfy an even-power expansion

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \cdots,$$

Richardson-type linear combinations of $Q_N, Q_{kN}, \dots$ cancel leading terms and increase the order of accuracy. This mechanism applies to polygonal approximations of π , chordal perimeter approximations, and geometric N-series acceleration.

The second mechanism is nonlinear residual-map iteration. In the Baker 4:3 trisection cascade, the active variable is not a resolution scale but the signed angular error, and the local map satisfies

$$e_{j+1} = -\frac{7}{384}e_j^3 + O(e_j^5).$$

Thus the cascade multiplies error order under iteration, producing

$$3 \longrightarrow 9 \longrightarrow 27 \longrightarrow 81 \longrightarrow \cdots.$$

The contribution of the note is therefore taxonomic rather than a claim of new Richardson theory. Scaling–cancellation is an order-increasing fixed-resolution mechanism; the Baker 4:3 cascade is an order-multiplying nonlinear refinement. Both suppress leading error terms by geometrically selected coefficients or scalings, but they act in different variables and belong to different convergence classes.

1 Introduction

Chord-based geometric approximation appears throughout classical and modern geometry. Archimedes’ polygonal approximation to π , chordal perimeter approximations for ellipses, and perimeter approximations for smooth convex curves all involve a discretization parameter N . As N grows, the approximation improves, and the leading error often has a predictable asymptotic structure.

A recurring form is

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \cdots,$$

where Q is the target geometric quantity and Q_N is its N -resolution approximation. The even powers arise from symmetry: local odd-order errors cancel, leaving a hierarchy in powers of N^{-2} .

The purpose of this paper is to isolate the elementary but useful consequence of this structure. If Q_N and Q_{kN} are available, then their leading errors differ by the factor k^{-2} . Therefore a specific

linear combination cancels the leading term. This is the geometric form of a Richardson-type acceleration mechanism.

The paper also clarifies what this mechanism is not. It is not a modular transformation, and it is not the same mechanism as the Baker 4:3 trisection cascade. The latter is a nonlinear residual-map iteration in the signed angular error. Its local law is cubic in the current error, not a fixed-resolution cancellation in N^{-2} . The distinction is important: scaling–cancellation increases the order of a fixed asymptotic expansion, whereas the Baker cascade multiplies the local error order under iteration.

2 Even-Power Asymptotic Structure

Let Q be a geometric quantity associated with a smooth curve, polygonal limit, or chord-based construction. Let Q_N be an approximation obtained from N uniform samples, chords, or subdivisions.

The following assumption records the asymptotic structure used throughout the paper.

Assumption 1 (Even-power chord expansion). *The approximation Q_N admits an asymptotic expansion of the form*

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + O(N^{-8}),$$

where the constants C_{2j} depend on the underlying geometry but not on N .

This assumption is natural for symmetric chord-based approximations. For example, if a smooth curve is sampled at uniform parameter spacing $h = 1/N$, then Taylor expansion of the local chord contribution produces powers of h . Symmetry under reversal of the local interval cancels the odd-order contributions, leaving even powers of h , equivalently even powers of N^{-1} .

Remark 1. *The paper does not require a universal proof that every chord-based approximation has such an expansion. The scaling–cancellation theorem is conditional on the displayed expansion. In applications, the expansion may be proved by Taylor expansion, Euler–Maclaurin analysis, or standard perimeter approximation estimates.*

3 The Scaling Law

The even-power expansion immediately gives a scaling law.

Proposition 1 (Scaling of the leading errors). *Assume*

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + O(N^{-8}).$$

For a fixed integer $k \geq 2$,

$$Q - Q_{kN} = \frac{C_2}{k^2} N^{-2} + \frac{C_4}{k^4} N^{-4} + \frac{C_6}{k^6} N^{-6} + O(N^{-8}).$$

Proof. Substitute kN for N in the asymptotic expansion:

$$Q - Q_{kN} = C_2 (kN)^{-2} + C_4 (kN)^{-4} + C_6 (kN)^{-6} + O((kN)^{-8}).$$

Since k is fixed, this becomes

$$Q - Q_{kN} = \frac{C_2}{k^2} N^{-2} + \frac{C_4}{k^4} N^{-4} + \frac{C_6}{k^6} N^{-6} + O(N^{-8}).$$

□

Thus the leading error coefficient is transformed by the predictable factor k^{-2} . This is the source of the cancellation formula.

4 The Scaling–Cancellation Principle

Theorem 1 (Scaling–cancellation principle). *Suppose*

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + O(N^{-6}).$$

For fixed $k \geq 2$, define

$$\widehat{Q}_{N,k} = Q_{kN} + \frac{Q_{kN} - Q_N}{k^2 - 1}.$$

Equivalently,

$$\widehat{Q}_{N,k} = \frac{k^2 Q_{kN} - Q_N}{k^2 - 1}.$$

Then

$$Q - \widehat{Q}_{N,k} = O(N^{-4}).$$

More precisely,

$$Q - \widehat{Q}_{N,k} = -\frac{C_4}{k^2} N^{-4} + O(N^{-6}).$$

Proof. From the expansion,

$$Q_N = Q - C_2 N^{-2} - C_4 N^{-4} + O(N^{-6}),$$

and

$$Q_{kN} = Q - \frac{C_2}{k^2} N^{-2} - \frac{C_4}{k^4} N^{-4} + O(N^{-6}).$$

Using

$$\widehat{Q}_{N,k} = \frac{k^2 Q_{kN} - Q_N}{k^2 - 1},$$

we compute

$$\widehat{Q}_{N,k} - Q = \frac{k^2(Q_{kN} - Q) - (Q_N - Q)}{k^2 - 1}.$$

Now

$$k^2(Q_{kN} - Q) = -C_2 N^{-2} - \frac{C_4}{k^2} N^{-4} + O(N^{-6}),$$

while

$$Q_N - Q = -C_2 N^{-2} - C_4 N^{-4} + O(N^{-6}).$$

Subtracting gives

$$k^2(Q_{kN} - Q) - (Q_N - Q) = C_4 \left(1 - \frac{1}{k^2}\right) N^{-4} + O(N^{-6}).$$

Dividing by $k^2 - 1$,

$$\widehat{Q}_{N,k} - Q = \frac{C_4}{k^2} N^{-4} + O(N^{-6}).$$

Therefore

$$Q - \widehat{Q}_{N,k} = -\frac{C_4}{k^2} N^{-4} + O(N^{-6}),$$

as claimed. □

Remark 2. *The combination*

$$\widehat{Q}_{N,k} = Q_{kN} + \frac{Q_{kN} - Q_N}{k^2 - 1}$$

uses only the two approximations Q_N and Q_{kN} . It does not require knowledge of C_2 . The coefficient C_2 is eliminated because its scaling factor k^{-2} is known.

5 Higher-Order Fixed-Scale Cancellation

The same idea extends to more than two scales. Suppose the full even-power expansion has the form

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \dots .$$

Choose distinct fixed scale factors

$$k_0, k_1, \dots, k_s, \quad k_j > 0,$$

and form a linear combination

$$\widehat{Q}_N^{(s)} = \sum_{j=0}^s w_j Q_{k_j N}.$$

To preserve the limiting value Q , the weights must satisfy

$$\sum_{j=0}^s w_j = 1.$$

To cancel the first s even-power error terms, require

$$\sum_{j=0}^s w_j k_j^{-2\ell} = 0, \quad \ell = 1, 2, \dots, s.$$

Theorem 2 (Higher-order fixed-scale cancellation). *Assume*

$$Q - Q_N = \sum_{\ell=1}^{s+1} C_{2\ell} N^{-2\ell} + O(N^{-2s-4}).$$

Let

$$k_0 = 1, \quad k_1, \dots, k_s$$

be distinct positive scale factors, and define

$$\widehat{Q}_N^{(s)} = \sum_{j=0}^s w_j Q_{k_j N}.$$

Suppose the weights satisfy

$$\sum_{j=0}^s w_j = 1$$

and

$$\sum_{j=0}^s w_j k_j^{-2\ell} = 0, \quad \ell = 1, 2, \dots, s.$$

Then

$$Q - \widehat{Q}_N^{(s)} = O(N^{-2s-2}).$$

For distinct values

$$k_0^{-2}, k_1^{-2}, \dots, k_s^{-2},$$

these conditions form a nonsingular Vandermonde system. Hence the weights exist and are unique.

6 Application to Polygonal Approximation of π

For an inscribed regular N -gon in the unit circle, the perimeter is

$$P_N = 2N \sin\left(\frac{\pi}{N}\right).$$

Using the sine expansion,

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + O(x^7),$$

we obtain

$$P_N = 2N \left[\frac{\pi}{N} - \frac{\pi^3}{6N^3} + \frac{\pi^5}{120N^5} + O(N^{-7}) \right].$$

Therefore

$$P_N = 2\pi - \frac{\pi^3}{3}N^{-2} + \frac{\pi^5}{60}N^{-4} + O(N^{-6}).$$

Thus

$$2\pi - P_N = \frac{\pi^3}{3}N^{-2} - \frac{\pi^5}{60}N^{-4} + O(N^{-6}).$$

Applying the scaling–cancellation theorem gives

$$\hat{P}_{N,k} = P_{kN} + \frac{P_{kN} - P_N}{k^2 - 1},$$

with

$$2\pi - \hat{P}_{N,k} = O(N^{-4}).$$

For $k = 3$,

$$\hat{P}_{N,3} = P_{3N} + \frac{P_{3N} - P_N}{8}.$$

Remark 3. *This construction does not prove a new value of π . It accelerates a known convergent polygonal approximation by cancelling its leading asymptotic error term.*

7 Application to Smooth Convex-Curve Perimeters

Let Γ be a smooth closed convex curve with perimeter P , and let P_N be a chordal perimeter approximation using N equal-parameter samples. Under standard smoothness and symmetry assumptions, one obtains an even-power expansion

$$P - P_N = C_2N^{-2} + C_4N^{-4} + O(N^{-6}).$$

The scaling–cancellation theorem gives

$$\hat{P}_{N,k} = P_{kN} + \frac{P_{kN} - P_N}{k^2 - 1},$$

and therefore

$$P - \hat{P}_{N,k} = O(N^{-4}).$$

For $k = 3$,

$$\hat{P}_{N,3} = P_{3N} + \frac{P_{3N} - P_N}{8}.$$

This is the same structural mechanism used in chordal ellipse-perimeter corrections. The method does not depend on the ellipse specifically; the ellipse is a particularly important example because its exact perimeter is non-elementary and classical approximations such as Ramanujan’s formula act on a different variable, namely eccentricity.

8 Comparison with the Baker 4:3 Trisection Cascade

The preceding applications are fixed-resolution approximations. Their active asymptotic variable is N^{-2} , and the improvement is obtained by combining approximants at N , kN , and possibly further fixed scales.

The Baker 4:3 trisection cascade belongs to a different class. There the active variable is not a discretization scale but the signed angular error

$$e_j = D_j - \frac{\theta}{3}.$$

For the Baker 4:3 branch, the local residual map has the form

$$e_{j+1} = -\frac{7}{384}e_j^3 + O(e_j^5).$$

After normalization, this becomes the degree-three normal form

$$\eta_{j+1} = -\eta_j^3 + O(\eta_j^5).$$

Thus the trisection cascade is not a direct instance of the scaling–cancellation theorem. It is a nonlinear residual-map iteration whose local effect is to multiply the current error order by three:

$$3 \longrightarrow 9 \longrightarrow 27 \longrightarrow 81 \longrightarrow \cdots.$$

The comparison is nevertheless useful. In both settings, geometry supplies a distinguished coefficient, scaling, or correction that suppresses a leading error contribution. But the mechanisms are logically distinct:

- scaling–cancellation is order-increasing cancellation in a fixed-resolution parameter;
- the Baker 4:3 cascade is order-multiplying nonlinear iteration in the signed error variable.

This distinction prevents the fixed N-series mechanism from being mistaken for a modular transformation or for the Baker cubic residual cascade.

9 Discussion

The scaling–cancellation mechanism has three ingredients:

- symmetry produces an even-power expansion;
- scaling gives predictable transformations of the asymptotic terms;
- linear combination cancels the leading error contribution.

This produces systematic order improvement without nonlinear iteration. In the simplest case, a pair of approximants Q_N and Q_{kN} cancels the leading N^{-2} term and produces fourth-order accuracy. With more fixed scales, successive even-power terms may be cancelled.

The mechanism is therefore best described as a Richardson-type geometric acceleration. It should be distinguished from super-exponential modular iteration, such as the arithmetic–geometric mean, and from nonlinear geometric residual cascades, such as the Baker 4:3 trisection construction.

10 Conclusion

We have formulated a scaling–cancellation principle governing chord-based geometric approximations with even-power asymptotic expansions. If

$$Q - Q_N = C_2 N^{-2} + C_4 N^{-4} + O(N^{-6}),$$

then

$$\widehat{Q}_{N,k} = Q_{kN} + \frac{Q_{kN} - Q_N}{k^2 - 1}$$

satisfies

$$Q - \widehat{Q}_{N,k} = O(N^{-4}).$$

The framework applies directly to polygonal approximations of π , chordal perimeter approximations for ellipses, and smooth convex-curve perimeter approximations. It also clarifies the relationship between geometric N-series acceleration and other refinement mechanisms.

The Baker 4:3 trisection cascade is not a scaling–cancellation operator. It is a separate cubic residual-map cascade. Together, these examples show two complementary forms of geometric error suppression: order-increasing cancellation in a resolution variable and order-multiplying iteration in an error variable.

Future work will extend the fixed-resolution scaling–cancellation framework to higher-order combinations, nonuniform chord distributions, and broader classes of smooth curves.

References

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